

Radha Mastandrea

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Positions

Schmidt AI in Science Postdoctoral Fellow

University of Chicago

Chicago, IL

Sep. 2025 - Present

My research leverages physics-aware machine learning to improve model-agnostic searches for new physics. My interests span a variety of signal signatures (such as new particles and hidden sectors) as well as subfields (including phenomenology and astronomy). Ultimately, I hope to show that when a search is built upon an neural network architecture that is customized to the expected physics or symmetries of the signal, what results is a highly powerful and interpretable new physics-detecting strategy.

Research Intern in Foundation Models for Science

Polymathic AI

NYC, NY

May. 2025 - Aug. 2025

During this internship, I explored whether multi-modal foundation models (i.e. large neural networks trained on diverse data formats) can outperform single-modal foundation models and non-foundation models on scientific tasks. Working in a small team, I developed a multi-model foundation model for astronomy that explicitly learned both single and cross-modal information. I also benchmarked the model on a variety of astronomical tasks such as galaxy morphology classification, redshift estimation, and similarity retrieval against other neural networks for astronomy. *Workshop paper in progress.*

Education

PhD in Physics

University of California, Berkeley

Berkeley, CA

Aug. 2021 - May 2025

Designated Emphasis in Computational and Data Science and Engineering

Thesis: Casting a Wide (Neural) Net: Using Machine Learning to Search for New Physics

Thesis Advisor: Ben Nachman

MPhil in Physics

University of Cambridge

Cambridge, UK

Oct. 2020 - Sep. 2021

Thesis: Investigating non-standard sources of parity violation at the LHC

Thesis Advisor: Christopher Lester

MASt in Physics

University of Cambridge

Cambridge, UK

Oct. 2019 - Jun. 2020

Distinguished Performance

Thesis: Search for new physics in $B_{(s)}^0 \rightarrow \mu^+ \mu^- \mu^+ \mu^-$ decays

Thesis Advisor: Valerie Gibson

B.S. in Physics

MIT

Cambridge, MA

Aug. 2015 - Jun. 2019

Thesis: Analyzing CMS Open Collider Data through Topic Modeling

Thesis Advisor: Jesse Thaler

Publications and Preprints

- [16] **Radha Mastandrea**, Shiyu Peng, Benjamin Rosser, and Matt LeBlanc. *Fast BIB simulation at a future Muon Collider with generative machine learning*. 2026. arXiv: [2609.12054](https://arxiv.org/abs/2609.12054) [physics.ins-det]. URL: <https://arxiv.org/abs/2609.12054>
- [15] Ranit Das, Marie Hein, Gregor Kasieczka, Michael Krämer, Lukas Lang, **Radha Mastandrea**, Louis Moureaux, Alexander Mück, and David Shih. “Kitchen Sink Anomaly Detection”. In: (Apr. 2026). arXiv: [2604.20965](https://arxiv.org/abs/2604.20965)

[hep-ph]

- [14] Rikab Gambhir and Radha Mastandrea. “Resummed distribution functions: making perturbation theory positive and normalized”. In: *JHEP* 06 (2026), p. 243. DOI: [10.1007/JHEP06\(2026\)243](https://doi.org/10.1007/JHEP06(2026)243). arXiv: [2512.04160](https://arxiv.org/abs/2512.04160) [hep-ph]
- [13] Chi Lung Cheng, Ranit Das, Runze Li, **Radha Mastandrea**, Vinicius Mikuni, Benjamin Nachman, David Shih, and Gup Singh. “Generator Based Inference (GBI)”. in: (May 2025). arXiv: [2506.00119](https://arxiv.org/abs/2506.00119) [hep-ph]
- [12] Rikab Gambhir, **Radha Mastandrea**, Benjamin Nachman, and Jesse Thaler. “Isolating Unisolated Upsilon with Anomaly Detection in CMS Open Data”. In: *Phys. Rev. Lett.* 135.2 (2025), p. 021902. DOI: [10.1103/vvv3-5kk1](https://doi.org/10.1103/vvv3-5kk1). arXiv: [2502.14036](https://arxiv.org/abs/2502.14036) [hep-ph]
- [11] **Radha Mastandrea**, Benjamin Nachman, and Tilman Plehn. “Constraining the Higgs potential with neural simulation-based inference for di-Higgs production”. In: *Phys. Rev. D* 110 (5 Sept. 2024), p. 056004. DOI: [10.1103/PhysRevD.110.056004](https://doi.org/10.1103/PhysRevD.110.056004). URL: <https://link.aps.org/doi/10.1103/PhysRevD.110.056004>
- [10] Kehang Bai, **Radha Mastandrea**, and Benjamin Nachman. “Non-resonant anomaly detection with background extrapolation”. In: *Journal of High Energy Physics* 2024.4 (Apr. 2024). ISSN: 1029-8479. DOI: [10.1007/jhep04\(2024\)059](https://doi.org/10.1007/jhep04(2024)059). URL: [http://dx.doi.org/10.1007/JHEP04\(2024\)059](http://dx.doi.org/10.1007/JHEP04(2024)059)
- [9] Tobias Golling, Samuel Klein, **Radha Mastandrea**, Benjamin Nachman, and John Andrew Raine. “Morphing one dataset into another with maximum likelihood estimation”. In: *Phys. Rev. D* 108 (9 Nov. 2023), p. 096018. DOI: [10.1103/PhysRevD.108.096018](https://doi.org/10.1103/PhysRevD.108.096018). URL: <https://link.aps.org/doi/10.1103/PhysRevD.108.096018>
- [8] Tobias Golling, Gregor Kasieczka, Claudius Krause, **Radha Mastandrea**, Benjamin Nachman, John Andrew Raine, Debajyoti Sengupta, David Shih, and Manuel Sommerhalder. “The interplay of machine learning-based resonant anomaly detection methods”. In: *The European Physical Journal C* 84.3 (Mar. 2024). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-024-12607-x](https://doi.org/10.1140/epjc/s10052-024-12607-x). URL: <http://dx.doi.org/10.1140/epjc/s10052-024-12607-x>
- [7] Tobias Golling, Samuel Klein, **Radha Mastandrea**, and Benjamin Nachman. “Flow-enhanced transportation for anomaly detection”. In: *Phys. Rev. D* 107.9 (2023), p. 096025. DOI: [10.1103/PhysRevD.107.096025](https://doi.org/10.1103/PhysRevD.107.096025). arXiv: [2212.11285](https://arxiv.org/abs/2212.11285) [hep-ph]
- [6] **Radha Mastandrea** and Benjamin Nachman. “Efficiently Moving Instead of Reweighting Collider Events with Machine Learning”. In: *36th Conference on Neural Information Processing Systems: Workshop on Machine Learning and the Physical Sciences*. Dec. 2022. arXiv: [2212.06155](https://arxiv.org/abs/2212.06155) [hep-ph]
- [5] Gregor Kasieczka, **Radha Mastandrea**, Vinicius Mikuni, Benjamin Nachman, Mariel Pettee, and David Shih. “Anomaly detection under coordinate transformations”. In: *Phys. Rev. D* 107.1 (2023), p. 015009. DOI: [10.1103/PhysRevD.107.015009](https://doi.org/10.1103/PhysRevD.107.015009). arXiv: [2209.06225](https://arxiv.org/abs/2209.06225) [hep-ph]
- [4] Barry M. Dillon, **Radha Mastandrea**, and Benjamin Nachman. “Self-supervised anomaly detection for new physics”. In: *Phys. Rev. D* 106.5 (2022), p. 056005. DOI: [10.1103/PhysRevD.106.056005](https://doi.org/10.1103/PhysRevD.106.056005). arXiv: [2205.10380](https://arxiv.org/abs/2205.10380) [hep-ph]
- [3] Christopher G. Lester, **Radha Mastandrea**, Daniel Noel, and Rupert Tombs. “Hunting for vampires and other unlikely forms of parity violation at the Large Hadron Collider”. In: *JHEP* 08 (2022), p. 231. DOI: [10.1007/JHEP08\(2022\)231](https://doi.org/10.1007/JHEP08(2022)231). arXiv: [2205.09876](https://arxiv.org/abs/2205.09876) [hep-ph]
- [2] Brian T Cook, Deborah F Woods, Jessica D Ruprecht, Jacob Varey, **Radha Mastandrea**, Kaylee de Soto, Jacob F Harburg, Umaa Rebbapragada, and Ashish A Mahabal. “Tracing Milky Way substructure with an RR Lyrae hierarchical clustering forest”. In: *Monthly Notices of the Royal Astronomical Society* (Apr. 2022). stac1007. ISSN: 0035-8711. DOI: [10.1093/mnras/stac1007](https://doi.org/10.1093/mnras/stac1007). eprint: <https://academic.oup.com/mnras/advance-article-pdf/doi/10.1093/mnras/stac1007/43400845/stac1007.pdf>. URL: <https://doi.org/10.1093/mnras/stac1007>
- [1] Patrick T. Komiske, **Radha Mastandrea**, Eric M. Metodiev, Preksha Naik, and Jesse Thaler. “Exploring the Space of Jets with CMS Open Data”. In: *Phys. Rev. D* 101.3 (2020), p. 034009. DOI: [10.1103/PhysRevD.101.034009](https://doi.org/10.1103/PhysRevD.101.034009). arXiv: [1908.08542](https://arxiv.org/abs/1908.08542) [hep-ph]

Talks

** denotes invited talks

ContrAst: Factorized Representation Learning for Multimodal Data

Open SkAI 2026, *SkAI Institute*

Sep. 2026

AI For Multimodal Science Workshop, *UC San Diego***

Jun. 2026

Generative AI for BIB Simulation

Artificial Intelligence for Muon Colliders Mini-Workshop, *Virtual***

Jun. 2026

IMCC Annual Meeting, *CERN***

Jun. 2026

Resummed Distribution Functions: making perturbation theory positive and normalized

ML4Jets 2026, *Hörsaalzentrum University of Vienna*

Sep. 2026

BOOST Physics workshop, *Jagiellonian University*

Jul. 2026

11th LCTP Spring Symposium: Theoretical Physics and AI, *University of Michigan***

May 2026

Isolating Unisolated Upsilons with Anomaly Detection in CMS Open Data

QCD@LHC, *Oxford University*

Oct. 2026

ML4Jets 2025, *Caltech*

Aug. 2025

AD4HEP, *Columbia University, Nevis Laboratories*

Jun. 2025

Constraining the Higgs Potential Shape with Machine Learning

University of Cincinnati HEP / Astrophysics Seminar, *University of Cincinnati***

Nov. 2025

Rutgers NHETC Seminar, *Rutgers University***

Dec. 2024

ATLAS ML Forum, *CERN*

Nov. 2024

CERN Theory ML Forum, *CERN*

Nov. 2024

CMS ML Forum, *CERN*

Nov. 2024

ML4Jets, *LPNHE*

Nov. 2024

DPF-PHENO, *University of Pittsburgh*

May 2024

A Survey of Machine Learning Methods for Anomaly Detection

Workshop on Machine Learning and High-Energy Physics, *Campus Akademie, Vienna*

Dec. 2023

The Interplay of Machine Learning–based Resonant Anomaly Detection Methods

LHCb RTA-DPA Meeting, *CERN*

Nov. 2024

IAIFI Journal Club, *MIT***

Apr. 2024

RWTH HEP-ML Club, *RWTH Aachen University* **

Jan. 2024

SLAC AI Seminar, *SLAC***

Dec. 2023

Hammers & Nails, *Ascona, Switzerland*

Oct. 2023

ML4Jets, *DESY*

Nov. 2023

FETA: Flow-Enhanced Transportation for Anomaly Detection

Fermilab AI Meetings, *Fermilab* **

Jul. 2023

APS April Meeting, *Minneapolis*

Apr. 2023

University College London HEP Seminar, *University College London*

Dec. 2022

Imperial College London HEP Seminar, *Imperial College London*

Dec. 2022

UC Berkeley 4D Seminars, *UC Berkeley*

Oct. 2022

HEPSim2Real: Creating background templates with normalizing flows

ML4Jets, *Rutgers University*

Nov. 2022

Using symmetries to build better latent spaces for dijet representation learning

APS April Meeting, *New York*

Apr. 2022

Exploring the Parity of the Quark-Sector SME with Madgraph

Fourth Summer School on the Lorentz- and CPT-violating Standard-Model Extension, *ICUSS*

May 2021

Analyzing CMS Open Collider Data through Topic Modeling

BOOST Physics Workshop, *MIT*

Jul. 2019

Testing Parameterized Theories of General Relativity using Gravitational Waves

APS New England Section Meeting, *URI*

Dec. 2017

Selected Posters

ContrAst: Factorized Representation Learning for Multimodal Data

Interdisciplinary Science summit, *Duke University*

Jun. 2026

SciFM 2026, *The University of Chicago*

May 2026

Honors and Awards

Aug. 2026	GDS Outstanding Dissertation Award
Apr. 2024	Berkeley Grad Slam Semi-Finalist
Aug. 2023 - May 2025	Templeton TEX Fellow
Apr. 2023	Citadel PhD Summit Attendee
Mar. 2023	APS GDS Impact Award
Mar. 2022 - Mar. 2023	APS DSECOP Fellow
Sep. 2021 - Aug. 2024	NSF Graduate Research Fellow
Oct. 2019 - Jul. 2021	Marshall Scholarship
Jun. 2019	Joel Matthew Orloff Award for Outstanding Service
Jun. 2019	Phi Beta Kappa
Jun. 2019	Sigma Pi Sigma
Oct. 2019	FUTURE of Physics @ Caltech participant
May. 2018 - Sep. 2018	Heising-Simons @ MIT Physics Research Fellow

Summer Schools

Jan. 2025	GGI Theoretical Lectures on Fundamental Interactions, <i>GGI</i>
Jun. 2024	TASI, <i>UC Boulder</i>
Aug. 2023	SLAC Summer Institute, <i>SLAC</i>
Jul. 2023	CMS Open Data Workshop, <i>Fermilab</i>

Technical Skills

Programming Python, C++, Mathematica, Matlab

HEP Softwares ROOT, MadGraph, Pythia, Delphes